THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene prevents contact with air/water.
2	**(a)** Y	Reacting with cold water indicates more reactivity.
3	**(c)** CuO and CO_2	Calcination: $CuCO_3 \longrightarrow CuO + CO_2$.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Oxide layer protects the metal.
5	**(c)** S	Each displaces the next, so S is lowest.
6	**(d)** copper	$CuO + H_2 \longrightarrow Cu + H_2O$.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Layers repel and crystal breaks.
8	**(a)** $FeSO_4$ and $CuSO_4$	Zn is more reactive than Fe and Cu but less than Mg.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene cuts off air and moisture, preventing the violent reaction.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true: Al_2O_3 reacts with acids and bases; utensils use depends on good conductivity of heat and malleability, so R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oxide is amphoteric (1); Zinc or aluminium (1).	1+1
12	Reduction of oxide is easier by carbon (1); So sulphides/carbonates are first converted to oxides (1).	1+1
13	Protective Al_2O_3 film (1); Film prevents contact with air/water (1).	1+1
14	Very exothermic, hydrogen catches fire (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	ZnO is the oxidising agent and C is the reducing agent (1); Redox, reduction of metal oxide with carbon (1); Zinc is moderately reactive, so carbon can reduce its oxide (1).	1+1+1
16	Salt water increases electrical conductivity (1); Faster electrochemical corrosion (1); Rust formation speeds up (1).	1+1+1
17	NaCl gives Na^+ and Cl^- ions (1); Sugar remains as neutral molecules (1); No ions in sugar solution (1).	1+1+1
	OR: A, B more reactive than Fe (1); C is between Fe and Cu (1); Order: (A, B) > Fe > C > Cu > D (1).	1+1+1
18	X is CuO, Y is copper (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1).	1+1+1
	OR: Oxides of Na, K, Ca dissolve (1); CuO is insoluble (1); Soluble ones form alkalis (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Copper (1). (ii) Hydrogen; it burns with a pop sound with a burning matchstick (1). (iii) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1). (iv) Mg > Zn > Fe > Cu (1); the faster the bubbling, the more reactive the metal is (1).	1+1+1+2
20	Purification of bauxite to Al_2O_3 (1); Molten alumina with cryolite (1); Cathode: $Al^{3+} + 3\,e^- \longrightarrow Al$ (1); Anode: $2\,O^{2-} \longrightarrow O_2 + 4\,e^-$; graphite burns (1); Cryolite lowers the temperature and raises conductivity (1).	1+1+1+1+1
	OR: Black layer turns brown (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1); CuO is the oxidising agent (1); H_2 is the reducing agent (1); Copper is regenerated (1).	1+1+1+1+1

Mistakes to avoid:

- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as Fe_2O_3; it is Fe_3O_4 and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.

- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $\mathrm{FeSO_4}$; a less reactive metal cannot displace a more reactive one.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Na_2O	Soluble basic oxides form alkalis.
2	**(b)** turns red litmus blue	NaOH is formed.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Sacrificial protection.
4	**(a)** MCl_3	M^{3+}.
5	**(d)** aluminium	Cryolite lowers its melting point.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Reactivity decides displacement.
7	**(d)** decreases	Copper from the anode dissolves into the electrolyte.
8	**(b)** tin and lead	Solder-type alloy.
9	**(c)** A is true but R is false.	Galvanisation is true, but zinc is *more* reactive than iron and protects it even if the coating is broken; R is false.
10	**(d)** A is false but R is true.	Graphite is a non-metal that conducts, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Silver is below hydrogen (1); Sodium or calcium displaces hydrogen from water (1).	1+1
12	Strong electrostatic forces (1); Displacement of layers brings like ions close, causing repulsion (1).	1+1
13	HNO_3 is a strong oxidising agent (1); It oxidises H_2 to water and itself forms nitrogen oxides (1).	1+1
14	Zinc is more reactive than tin and dissolves in food acid (1); Zinc salts are harmful (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Copper reacts with CO_2 and moisture forming basic copper carbonate (1); Acids in lemon/tamarind react with the green layer (1); The basic carbonate dissolves leaving shiny copper (1).	1+1+1
16	Roasting in air gives copper(I) oxide (1); Reduction by self-reduction with Cu_2S (1); Electrolytic refining gives pure copper (1).	1+1+1
17	Brass is harder (1); More corrosion-resistant (1); Easily cast and has golden colour (1).	1+1+1
	OR: X is sodium (1); $4\,Na + O_2 \longrightarrow 2\,Na_2O$ (1); $Na_2O + H_2O \longrightarrow 2\,NaOH$ (1).	1+1+1
18	Ag and Au are least reactive (1); They do not react with oxygen, water (1); Sodium is highly reactive and forms stable compounds (1).	1+1+1
	OR: Foil: malleability (1); Wires: conductivity and ductility (1); Aircraft: light weight when alloyed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Roasting (1). (ii) Calcination (1). (iii) $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$ (1). (iv) $ZnO + C \longrightarrow Zn + CO$ (1); zinc lies in the middle of the reactivity series, so carbon can reduce its oxide (electrolysis is needed only for highly reactive metals like Na, Ca, Al) (1).	1+1+1+2
20	X: 2,8,8,2; Cl: 2,8,7 (1); X loses 2 electrons, each Cl gains 1 (1); Formula XCl_2 = $CaCl_2$ (1); Ionic, high melting point (1); Soluble in water, conducts when molten (1).	1+1+1+1+1
	OR: Ore is a profitable mineral (1); Roasting in excess air, calcination in limited air (1); Amalgam contains mercury (1); Galvanisation prevents corrosion with zinc (1); Ionic: transfer of electrons, covalent: sharing (1).	1+1+1+1+1

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- Writing the product of iron with steam as Fe_2O_3; it is Fe_3O_4 and hydrogen gas is released.

- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.
- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $\mathrm{FeSO_4}$; a less reactive metal cannot displace a more reactive one.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** zinc blocks on ship hull	Zinc corrodes first.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Delocalised electrons exist.
3	**(c)** charge +2, formula MgO	Mg (2,8,2) loses 2 electrons.
4	**(c)** roasting -> reduction -> refining	Sulphide ores are roasted first.
5	**(c)** A is true but R is false.	Calcium is denser; hydrogen bubbles make it float. R false.
6	**(c)** M_2S_3	M^{3+} and S^{2-}: 2 x 3 = 3 x 2.
7	**(a)** a sulphide	Sulphide ores need excess air.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Alloyed for hardness.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Electropositive means forming cations by loss of electrons, which R states.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; conduction in the molten state occurs because ions become free to move, not because of the high melting point.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	M is magnesium (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1).	1+1
12	$Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ (1); Used to weld railway tracks and cracked machine parts (1).	1+1
13	Anode: chlorine (1); Cathode: sodium (1).	1+1
14	(i) Yes, Cu is more reactive than Ag (1); (ii) No, Ag is less reactive (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fe: reduction with carbon/CO, medium reactivity (1); Na: electrolysis of molten NaCl, high reactivity (1); Carbon cannot reduce Na^+ (1).	1+1+1
16	Mixture shows properties of both (1); Heating gives a compound FeS (1); Magnet does not attract it (1).	1+1+1
17	$Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); Blue colour of the solution fades (1); Reddish-brown copper is deposited on the zinc (1).	1+1+1
	OR: Alumina melts at very high temperature (1); Cryolite lowers the melting point and increases conductivity (1); Graphite electrodes carry current (1).	1+1+1
18	Anodising: make Al the anode in dilute H_2SO_4 (1); Al_2O_3 layer becomes thicker (1); It resists corrosion and can be dyed (1).	1+1+1
	OR: Highly reactive: electrolysis (1); Moderately reactive: reduction by carbon (1); Least reactive: heating or found free (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Two (electronic configuration 2, 8, 2) (1). (ii) $MgCl_2$ (1). (iii) Ionic bond (1). (iv) In the solid the ions are held in fixed positions and cannot move (1); in the molten state or solution the ions are free to move and carry charge (1).	1+1+1+2
20	(a) no rust, no water (1); (b) no rust, no oxygen (1); (c) rust (1); Rusting needs both oxygen and moisture (1); Prevention: paint, oil, galvanising, alloying (1).	1+1+1+1+1
	OR: Reactive metals form stable compounds (1); $ZnCO_3 \longrightarrow ZnO + CO_2$ (calcination) (1); $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$ (roasting) (1); $ZnO + C \longrightarrow Zn + CO$ (1); Roasting/calcination makes oxide which is reduced (1).	1+1+1+1+1

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